THE UNITED REPUBLIC OF TANZANIA
PRIME MINISTER'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
MPWAPWA DISTRICT COUNCIL

FORM TWO PRE MOCK II EXAMINATION
032 – CHEMISTRY

TIME: 2:00 HRS **MARCH 2026**

INSTRUCTIONS

1. This paper consists of three sections A, B and C with total of **ten** Questions.
2. Section A and C carries **15** Marks and section B carries **70** marks.
3. All writings must be **Blue** or **Black ink** except for diagrams which must be in **pencil.**
4. The following atomic masses may be used**; H = 1, C = 12, N = 14, O = 16, S = 32, Cl = 35.5, K = 39**

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SECTION A (15 Marks)

Answer All Questions

1. For each of the items (i) – (x), choose the correct answer from among the given alternatives and write its letter in the box provided. **(10 Marks)**

i. Which of the following is a compound?
A. Helium (He) B. Hydrogen (H_2) C. Carbon dioxide (CO_2) D. Neon (Ne)

ii. Your brother was given a syrup with a prescription that IT SHOULD BE WELL SHAKEN BEFORE USE. This is because the syrup is a
A. Clear solution B. Suspension C. Homogenous mixture D. Filtrate

iii. The components of a mixture of iron and sand can be separated by
A. Boiling B. Chemical reaction C. Magnetic attraction D. Filtration

iv. Your chemistry teacher denied a student to enter the laboratory because of a floppy clothing. This is because the cloth may
A. Be blown by wind B. Affect the measurements
C. Catch fire or cause one to fall D. Cause poor ventilation in the body

v. Imagine that, you are playing a chemistry related trivia game with your friends. If you are to choose one term which does not relate to any recognized branches of chemistry among the following, which term could you choose?
A. Organic chemistry B. Arithmetic chemistry
C. Inorganic chemistry D. Physical chemistry

vi. If a form two aspires to be a professional farmer, which among the following subjects a student should be advised to focus on in order to develop the necessary competences?
A. Geography, Physics and Chemistry B. Biology, Physics and Mathematics C. Agriculture, Chemistry and Biology D. Chemistry, Physics and Biology

vii. Why non – luminous flame is the most applicable flame for heating and welding purposes:
A. It is very noise B. It is very hot C. It has no soot D. It has three zones

viii. The fallowing apparatus is used to keep test tubes in the laboratory

A. Test tube holder B. Test tube rack C. Test tube brush D. Basin for test tubes

ix. Mr. Mwaisa coated the roofs of his house with zinc in order to protect it from rusting. What was the process done by Mr. Mwaisa?
A. Painting B. Galvanization C. Oiling D. Alloying

x. When a chemist studies a substance, he/she is interested in its
A. Shape B. Properties C. Smell D. Force of attraction

2. Match the descriptions in LIST A with the correct techniques for separating mixtures in LIST B by writing the letter of the correct response in the table provided. **(05 Marks)**

LIST A	**LIST B**
i. Extraction of 5 g of table salt dissolved in 10 cm^3 of salt solution. ii. Separation of cooking oil from a mixture of water and cooking oil. iii. Collecting 50 cm_3 of pure water from 300 L copper (II) sulphate solution. iv. Extraction of flower pigment from hibiscus flower using 2.5cm^3 of ethanol. v. Separating serum from suspension of human blood	A. Funnel separation B. Solvent extraction C. Evaporation D. Fractional distillation E. Centrifugation F. Simple distillation G. Paper chromatography

SECTION B (70 Marks)
Answer All Questions

3. (a) Some of elements have their symbols completely different from their common English names.

(i) Where are their symbols derived from? **(01 mark)**
(ii) Use the table below to list three elements of this kind **(03 Marks)**

ENGLISH NAME	**LATIN NAME**

(b) The teacher told Ndahogora to list two elements which he knows. Ndahogora wrote Al and O. Why did he use letters instead of writing their full names? Give three (3) reasons **(03 Marks)**

(c) Briefly give reason for each of the following laboratory rules **(03 Marks)**

(i) Do not return unused chemicals in their original container, why?

(ii) Open the windows while you are working in laboratory, why?

(iii) Laboratory doors and windows must open outwards, why?

4. (a) The figure below is a part of the modern periodic table which includes the first 20 elements **(08 Marks)**

Q		D					C
H	M					L	

(i) Mention the chemical symbols of the elements represented by letters "Q" and "L"

(ii) Write the electronic configuration of the elements represented by the Letters "M" and "C"

(iii) Which letter represent element that belong to group III?

(iv) Which letters represent elements that belongs to the same group?

(v) Which letters represent elements that belongs to the same period?

(b) Using clear statement distinguish between Mendeleev's periodic law from Modern periodic law **(02 Marks)**

5. (a) Draw the structure of an atom according to the: (*label it if possible*) **(04 Marks)**
   i. Dalton's Atomic Theory
   ii. Rutherford Atomic Model

(b) Many scientists explained about the structure of an atom including John Dalton. State four postulates of Dalton's Atomic Theory. **(04 Marks)**

(c) State two components of Nucleons. **(02 Marks)**

6. (a) State the number of protons, neutrons and electrons in each of the following atoms. **(06 Marks)**
   (i) $_{13}^{27}Al$

(ii) $_{92}^{235}U$

$_{13}^{27}Al$	$_{92}^{235}U$
Protons =	Protons =
Neutrons =	Neutrons =
Electrons =	Electrons =

(b) Mention two application of isotopes **(04 Marks)**

7. (a) What do you understand by these terms
   i. Empirical formula
   ii. Molecular formula

(b) A certain organic compound composed of 26.7% carbon, 2.2% hydrogen and the rest is oxygen. If its relative atomic mass is 90. Calculate it's;

(i) Percentage of Oxygen
(ii) Empirical formula
(iii) Molecular formula

8. (a) How does chemistry contribute to the development of today's modern society. Give five points **(05 Marks)**

(b) Since chemistry is the science that involves experiments and practical works, chemists have to acquire scientific skills in order to be successful in obtaining facts and verifying them. Mention Five (5) skills to be acquired by chemists. **(05 Marks)**

9. (a) Ten students from Mgebuka secondary school were doing an experiment in the school laboratory with all windows closed in such a way that low circulation of air occurred in the room. Some containers of chemicals such as concentrated acid was giving out fumes hence made the students to suffocate. If you were among the students in the school laboratory, what are two (2) things that you would have done in order to rescue yourself and others from suffering the same problem as those occurred to the students? **(04 Marks)**

(b) The study of Chemistry provides chemists with the knowledge and skills that are required in their future career opportunities. Mention any six professions that chemists can acquire in the future. **(03 Marks)**

(c) A good chemistry laboratory has various qualities and characteristics. As a form two student from Mgebuka secondary school Mention any three qualities of a good chemistry laboratory you know. **(03 Marks)**

SECTION C (15 Marks)
Answer Question 10

10. (a)Explain the meaning of the following terms
(i) Oxidation state (ii)
Valency

(b) Find the oxidation state of the following underlined elements
(i) $\underline{C}O_2$
(ii) $\underline{Cr}_2O_7^{2-}$
(iii)$\underline{S}O_3^{2-}$ (iv)
$K\underline{Cl}O_3$
(v) $\underline{N}H_4^+$

(c) Copper exists naturally as two isotopes, copper-63 ($^{63}_{29}Cu$)and copper-65, ($^{65}_{29}Cu$). The relative atomic mass of copper is 63.5. Calculate the percentage abundance of copper-63 and Copper-65.